

1. List all 6 phases of SDLC in order with one line about each.

The SDLC provides a structured framework for building high-quality software through the following six stages:

1. **Requirement Analysis:** Gathering detailed functional and technical needs from stakeholders to define the scope of the project.
2. **Design:** Creating the architectural blueprints, data models, and user interface layouts based on the requirements.
3. **Implementation (Coding):** Translating the design documents into actual source code using specific programming languages.
4. **Testing:** Identifying and fixing defects to ensure the software is reliable and meets the original requirements.
5. **Deployment:** Releasing the finalized software into a production environment where users can access it.
6. **Maintenance:** Providing ongoing support, bug fixes, and updates to ensure the software remains functional and relevant.

2. Write 3 differences between Agile and Waterfall models.

Feature	Waterfall Model	Agile Model
Project Structure	Follows a linear, sequential path where one phase must end before the next begins.	Follows an iterative approach where work is broken into small, repeatable cycles.
Requirements	Requirements are fixed at the start of the project and are difficult to change later.	Requirements are flexible and evolve based on continuous feedback and changing needs.

Feature	Waterfall Model	Agile Model
Delivery & Feedback	The final product is delivered at the end of the cycle, meaning feedback comes late.	Features are delivered in small increments, allowing for frequent testing and feedback.

3.Name the 3 main Scrum roles and write one line about the responsibility of each.

Scrum is a framework for Agile development that utilizes three specific roles to manage productivity:

- **Product Owner:** Responsible for maximizing the value of the product and managing the Product Backlog to ensure the team works on the most important features.
- **Scrum Master:** Acts as a facilitator who helps the team understand Scrum theory and removes any obstacles hindering their progress.
- **Developers:** The professionals who do the actual work of building a usable increment of the product during every Sprint.

4. What is a Sprint in Scrum? Mention its typical duration.

A **Sprint** is a fixed, time-boxed period during which a specific set of work is completed and made ready for review. It is the heartbeat of the Scrum framework.

- **Typical Duration:** A Sprint usually lasts between **1 to 4 weeks**, with 2 weeks being the most common industry standard.

5. What do these Jira board columns mean: To Do, In Progress, Done, Backlog?

Jira is utilized as a project tracking tool to monitor workflows and tasks. The standard columns represent the following stages of a task:

- **Backlog:** A list of all potential tasks, features, or bugs that need to be addressed but are not yet scheduled for active work.
- **To Do:** Tasks that have been selected for the current cycle (Sprint) and are waiting for a developer to start working on them.
- **In Progress:** Tasks that are currently being actively developed or addressed by a team member.
- **Done:** Tasks that have successfully passed all development, testing, and review criteria and are considered complete.